

ROADBLOCKS TO ZERO-ENERGY HOMES

In order to reach zero energy, a new home will require at least \$20,000 worth of PV capacity just to offset the consumption of miscellaneous devices!

BY RICH BROWN
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On the whole, houses in the United States are better insulated and have more efficient HVAC and domestic hot water systems than ever before. This is especially true for homes built to Energy Star and other efficiency and green building standards. But as homes become more efficient at heating, cooling, and making hot water, other energy uses in the home that had largely been ignored in the past come to the forefront.

Today, major appliances use approximately 25%, lighting uses approximately 20%, and miscellaneous equipment uses about 15% of a home's electricity. Nationwide, these end uses account for about 40% of total residential primary (or "source") energy consumption. Moreover, miscellaneous energy is the fastest-growing end use in U.S. homes; it is projected to more than double in the next 20 years. This is particularly true in new homes, where increasing floor area and added amenities require more electricity for lighting, electronics, and other plug load devices.

And while new homes have long been the subject of programs and policies to reduce energy use, these programs have focused almost exclusively on traditional end uses. (Notable exceptions are California's Title 24 building standard, which includes requirements for lighting efficiency, and the Energy Star New Home program, which requires some efficient appliances and lighting.) While these programs have succeeded in reduc-

ing traditional end uses, little progress has been made in reducing the other end uses in new homes. These miscellaneous loads are making it much more difficult to reach the goal of building affordable zero-energy homes. In order to reach zero energy, the average new home would require \$20,000–\$30,000 worth of PV capacity just to offset the consumption of miscellaneous devices!

Whole-House Monitoring

To learn more about the consumption of "other" end uses (appliances, lighting, electronics, and miscellaneous equipment) in new homes, we have

compiled the results of several studies of home energy use.

Some of the studies we looked at involved the long-term metering of electrical consumption at the whole-house level and the submetering of select major end uses. These end uses included space heating, space cooling, water heating, and sometimes major appliances, such as the refrigerator or the clothes dryer. (Table 1 describes the low-energy homes that were monitored.)

Civano. A development in Tucson, Arizona, Civano is designed to follow sustainability principles. Homes in the development were designed to consume 40%–50% less energy for heating, cooling, and water heating than homes meeting



BILL RITTELMANN

(left to right) Bill Rittelmann, Gale Proski-Marsland, and Bill Webber stand on a deck atop Webber's home at the Civano development. Rittelmann, of IBACOS, was studying the energy use of the efficient home and SolarBuilt, LLC, architect Proski-Marsland is the home's designer.

the 1993 Model Energy Code. Civano homes also have efficient appliances and lighting and solar water-heating systems, and some have roof-mounted PV systems. Integrated Building and Construction Solutions (IBACOS) helped to design the pilot phase of the Civano development, and conducted long-term monitoring on 20 homes in the pilot phase. Eight homes provided data that was adequate for this analysis. The homes were constructed in 2000 and 2001 and were monitored for a two-year period after occupancy. We report only one year of data here—calendar year 2001 in most cases. Energy measurements were conducted using current transducers at the circuit panel; current measurements were converted to power using a one-time voltage measurement. This method was found to be accurate to within 1% of the actual utility bills for a test house.

Aspen Homes. As part of DOE's Building America program, IBACOS also assisted with the design and monitoring of a low-energy home in Loveland, Colorado (built by Aspen Homes). This home was designed to consume 60% less energy than a standard home. It is highly insulated and airtight, and it features a 2 kW PV system, solar hot water, solar-assisted gas-fired boiler heating, a thermal wall assembly, high-performance lighting, and high-efficiency appliances. The home was constructed in 2004 and monitored through 2005 (we report data for calendar year 2005 in this article). The monitoring method was similar to that used for the Civano homes, but the submetering was performed at a much more detailed level, which allowed the disaggregation of consumption by device and electrical circuit type.

Tucson ZEH. Another Building America demonstration home was the Zero Energy Home (ZEH) at Armory Park del Sol in Tucson, Arizona. The National Association of Home Builders

(NAHB) Research Center assisted with the design and monitoring of this unoccupied home. It was designed to consume 45% less energy than the builder's standard design, which is itself designed to perform 50% better than the Model Energy Code (that is, the Tucson ZEH was designed to use a total of 72% less energy than the Model Energy Code). Design features include a highly insulated envelope, a solar space- and water-heating system with tankless electric water-heating backup, a high-efficiency air conditioner, pin-based fluorescent lighting fixtures, Energy Star-compliant major appliances, and a 4.2 kW PV array.

at "other" electricity consumption. The sample covered a range of construction vintages, including homes built in the mid- to late 1990s. This made it possible to determine whether new homes differ from older homes in their consumption for "other" end uses. The new homes in this study are broadly representative of building practices in Florida at that time and are not skewed toward low-energy designs.

Device Level Surveys and Metering

The second type of study we considered collected data on energy consumption at the level of the individual device. Rich Brown and Greg Homan, researchers at Lawrence Berkeley National Laboratory (LBNL), inventoried all the electrical equipment in 13 new, unoccupied homes in California. Brown performed spot metering of plug-in devices to measure their power consumption in low-power modes (that is, in all operating modes other than active—for

example, off or standby). Annual energy use was estimated using typical usage patterns and active-mode power use from other studies.

This sample of homes represents energy consumed by builder-installed devices (rather than the full complement of devices present in occupied homes). Some of the homes were sales models that included a few additional devices that would normally be found in an occupied home. The sample includes seven floor plans from four different housing developments. The homes represent a range of efficiency levels, from typical new construction in California to low-energy homes with solar PV systems.

How Much Do "Others" Use?

The conclusions in this article are based on a sample of approximately 25 new homes (Civano homes, Aspen

Table 1. Characteristics of Monitored Low-Energy Homes

House	Floor Area (ft ²)	Number of Occupants	Heating Type	Backup Water-Heating Fuel
Civano 1	1,556	1	Gas furnace	Gas
Civano 2	2,042	2	Heat pump	Electric
Civano 3	1,566	3	Heat pump	Electric
Civano 4	1,280	3	Fan coil from gas water heater	Gas
Civano 5	1,834	2	Gas furnace	Gas
Civano 6	1,834	2	Heat pump	Electric
Civano 7	1,834	3	Heat pump	Electric
Civano 8	1,227	2	Heat pump	Electric
Aspen	3,453	3	Gas boiler	Gas
Tucson ZEH	1,718	not known	Electric	Electric

The home was built in 2003 and monitored through 2005 (we report data for calendar year 2005 in this article).

Florida. Danny Parker of the Florida Solar Energy Center (FSEC) monitored approximately 200 existing homes in central Florida during 1999. The homes represented a broad cross section of building designs, heating fuel, and equipment and appliances. Only homes without swimming pools or spas, and where the energy use of the clothes dryer was measured, are included in the sample set used in this article. Total electricity use was metered in all the homes by monitoring incoming electrical service. Major end uses were also recorded in each home on a 15-minute basis. These end uses included space heating; space cooling; water heating; and either the clothes dryer, or the range. All the submetered end uses were subtracted from the total to arrive

home, Tucson ZEH, and 13 California homes, all built since 2000) in the western United States and approximately 200 homes built before 2000 in Florida. It should be noted that the sample of new homes represented by these studies is generally skewed toward energy-efficient and low-energy designs, because these tend to be the homes that have been most intensively monitored. The detailed submetering needed to measure consumption for appliances, lighting, and miscellaneous loads is expensive and has generally not been conducted on large samples of “average” new homes (the Florida study being a notable exception). The studies are also heavily skewed toward hot climates (Arizona, Florida, California), both because this is where a lot of houses are being built and because low-energy homes using solar energy systems are being tested in these climates first. For this reason, the houses described here should not be considered representative of national building practices. But they provide anecdotal evidence about a certain segment of the housing market.

Low-Energy Homes

On average, the low-energy homes monitored use about 8,400 kWh per year of electricity, of which “other” end uses account for 46%–88%, with an average of 65% (see Figure 1). Consumption for “other” end uses ranges from 3,200 to 10,000 kWh per year, with an average of 5,400 kWh per year. For comparison, the average existing house in the Mountain census division (where all the monitored homes are located) consumes about 9,900 kWh per year, of which “other” end uses account for



This home was built by KG&E Homes and is a part of the Community of Civano. Homes in Civano were designed and built for optimal solar gain in the winter and shading in the summer.

BILL RITTELMANN

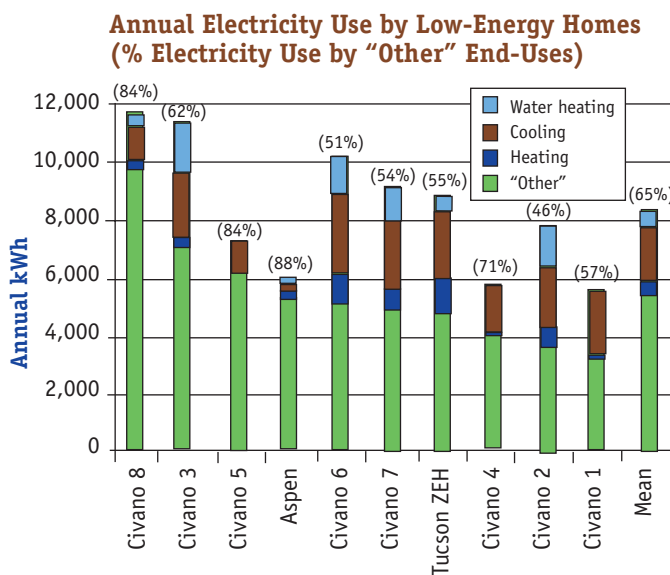


Figure 1. On average, the low-energy homes that were monitored use about 8,400 kWh annually, of which “other” end uses account for 46%–88%, with an average of 65%.

about 6,900 kWh per year, or 70% of whole-house consumption. (The end use consumption for existing homes is estimated using a regression-based bill disaggregation method, and is therefore not exactly comparable to the end use metered data presented here. Nevertheless it is useful as a general comparison.) Because most of these houses are located in Tucson, Arizona, a better comparison might be the average residential customer in the Tucson Electric Power service territory, which consumes about 20% more electricity—10,200 kWh per year.

Because of the hot climate in Tucson, the low-energy homes have large cooling and relatively small heating loads. For all the homes, cooling is the second largest end use after “other.” Some of the variation in “other” consumption is attributable to differences in house size. The rest is attributable to differences in the lifestyle of the occupants and the type of electrical devices they own. For instance, the occupants of one Civano house report doing a lot of laundry, which drives up their appliance use. Occupants of another house (Civano 8) have a lot of electronic and audio equipment, which may help to explain why their house has the highest “other” consumption of all the houses studied.

Although all of these houses have energy-efficient hard-wired lighting and Energy Star-compliant appliances wherever possible, their consumption for “other” end uses is only about 20% less than “other” consumption for the average home in that region.

More detailed end use metering in the Aspen house allows us to disaggregate the “other” end uses to see which components predominate. Plug circuits, at 2,439 kWh per

year, are the largest end use, followed by lighting and appliances. Lighting consumption is around 2,000 kWh per year, including some portable light fixtures. Appliance consumption is around 1,500 kWh per year, although a freezer in the garage is included in the plug consumption. With these adjustments, plug loads, lighting, and appliances all consume about the same amount of energy in the Aspen house.

Existing Homes in Florida

The metering study of homes in Florida provides a larger data set to in-

investigate whether “other” electricity consumption differs between new and existing homes. (“New” in this case means built in the 1990s.) “Other” electricity consumption varies significantly between houses, but we found no statistical correlation with the age of the house. Again, the mean consumption is around 6,000 kWh per year, which is similar to consumption for the low-energy homes described above. Although absolute consumption for “other” does not appear to vary with the age of the house, “other” end uses may comprise a larger fraction of whole-house consumption in new homes. This is because the traditional end uses have been reduced in new homes through improved building and equipment standards.

On the other hand, there does appear to be a statistical relationship (significant at the 99% level) between “other” electricity consumption and floor area. The results seem counter intuitive. There is no discernible difference between vintages as far as “other” electricity consumption is concerned, whereas one would expect new homes to have higher “other” consumption due to their larger floor area. A possible explanation for these findings is that the effect of larger floor area (leading to higher saturation of electrical devices) has been largely offset by gains in efficiency for appliances and other equipment in the “other” end uses.

Builder-Installed Devices

As one would expect, the study of 13 new unoccupied California homes

Annual Electricity Use by Builder-Installed Equipment in New California Homes, by Operating Mode

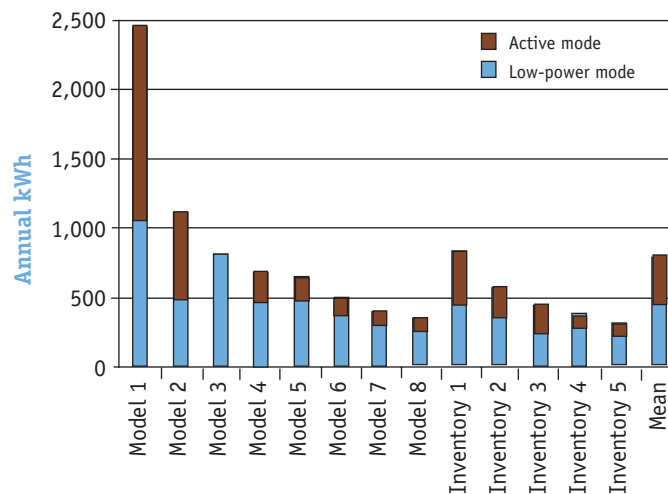


Figure 2. The consumption by these builder-installed devices is about 800 kWh per year on average.

Annual Electricity Use by Builder-Installed Equipment in New California Homes, by Equipment Type

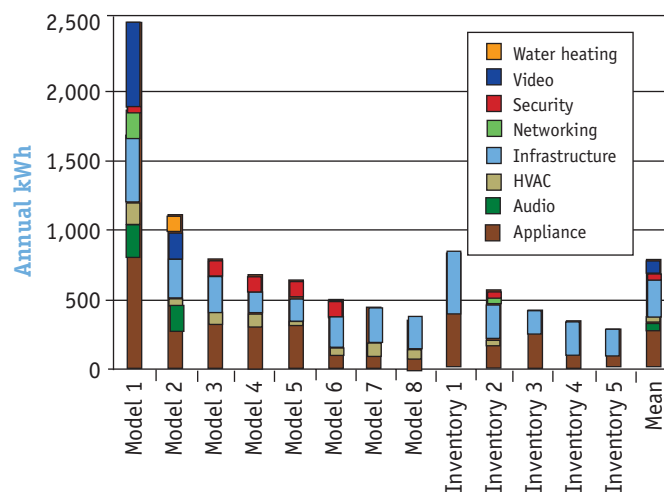


Figure 3. Devices that help to account for Model 1’s high electricity consumption include video (two large-screen televisions); networking (a structured wiring panel that draws 20 watts continuously to power three video security cameras and an Internet router); infrastructure (a larger floor area leads to more GFCI outlets, smoke alarms, garage door openers, bathroom fans, and so on); and appliances (a washer/dryer and very large refrigerator).

found fewer devices than are normally found in occupied homes. This helps to account for the lower electricity consumption in unoccupied new homes (compared to consumption in the whole-house occupied-home studies cited above). The consumption by these

builder-installed devices is about 800 kWh per year on average (see Figure 2). Note that these results include only energy use for nonlighting equipment. One of the main purposes of this study was to determine how much electricity is consumed in low-power versus active modes. The results in Figure 2 indicate that low-power modes account for about half of the electricity consumed by builder-installed devices. The mean low-power consumption (approximately 440 kWh per year) translates to a continuous standby power draw of about 50 watts.

Looking in more detail at the specific types of devices that are found in a typical new home, we found that several devices show higher than expected standby power. These include gas fireplaces (5 watts for a typical unit), the structured wiring panel and power supply (16 watts), and the garage door opener (5.4 watts). Because this is a model home, it contains a few items, such as a television, that would not normally be present in an unoccupied new home. Nevertheless, it is clear that the electricity use of builder-installed devices will need to be addressed in future efficiency programs.

The inventory homes include only the equipment and devices that would normally be in the home when it was sold to a buyer. Model homes include additional devices (for example, appliances and video and audio equipment) used to decorate the sales model for

prospective home buyers. As expected, the model homes have somewhat higher consumption, particularly houses 1 and 2.

To learn which devices account for the variation between homes, we looked at annual electricity consumption disaggregated by type of device (see Figure 3). Devices that help to account for Model 1's high electricity consumption include video (two large-screen televisions); networking (a structured wiring panel that draws 20 watts continuously to power three video security cameras and an Internet router); infrastructure (a larger floor area leads to more GFCI outlets, smoke alarms, garage door openers, bathroom fans, and so on); and appliances (a washer/dryer and very large refrigerator). Model 2 has higher than average energy use mainly in that it includes video (a television) and water heating (standby power for an instantaneous gas water heater).

Note that consumption data from the device-level surveys are not directly comparable to consumption data from the whole-house studies presented above. There are three reasons why. First, energy consumption in the surveys is a bottom-up estimate based on measured power levels and imputed usage, rather than a long-term direct measurement. Second, lighting is not included in the consumption estimates for the builder-installed devices. And third, these homes contained very few devices that would normally be supplied by the occupants. Nevertheless, the data do suggest how much "other" end use consumption can be addressed through energy-efficiency efforts targeted at home builders.

A Comprehensive Picture

By combining the results of several metering studies of new homes, we have tried to assemble a more comprehensive picture of "other" end use electricity consumption in new homes. We found



Civano homes were built for comfort and for aesthetic appeal as well as for energy and water efficiency.

that even in low-energy houses, "other" end uses account for 60% of electricity consumption on average! For one low-energy house with more detailed submetered data, "other" electricity consumption is fairly evenly split among plug loads, lighting, and appliances.

The available evidence also suggests that "other" electricity consumption is not significantly different in new and existing homes. The reason for this is not clear, but one hypothesis is that two counteracting forces are at play. New homes have larger floor areas and more "other" devices, which drives energy use up, but the devices in new homes tend to be newer and more efficient (thanks to improved equipment and building standards), which drives energy use down.

Finally, it appears that builder-installed "other" equipment contributes substantially to "other" electricity use. This suggests that devices not tradi-

tionally addressed in energy efficiency programs (such as garage door openers, structured wiring systems, smoke alarms, ventilation fans, and even GFCI outlets) might be worth upgrading through residential new-construction programs.

These findings suggest that electricity consumption for "other" end uses is still too great to make possible cost-effective zero-energy homes. This is especially true when electricity use must be offset with expensive PV power.

Clearly further research is needed in this important area of energy use. We need to learn more about energy consumption for "other" end uses, and we also need to develop more effective energy-saving strategies for these end uses, particularly for devices that have not been addressed by labeling programs or building codes. The results from the low-energy home meter-

ing study indicate that efficient appliances and lighting do not significantly reduce "other" energy use, a result that certainly raises questions about how we are currently addressing these end uses. Ultimately, most of this energy use is attributable to lifestyle decisions on the part of the occupants. They need better information about the consequences of these decisions, and about the options for reducing "other" energy use.



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